

New Theoretical Framework Challenges Particle Dark Matter Paradigm: "Chrono-Acoustic Brane Mechanics" Predicts Falsifiable Quantum Flicker

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Independent research collective *AINUMPSA Collective*, operating under the *HOMOMACHINE Project*, has published a provocative preprint on the Zenodo repository titled

"Chrono-Acoustic Brane Mechanics (CABM) v0.2: Unifying Dark Matter and Dark Energy via Superluminal Ontological Tension and Novikov Self-Consistency."

The paper, authored by theoretical researcher Chris Numpsa in collaboration with an advanced generative AI engine, proposes a total elimination of the dark matter particle paradigm (such as WIMPs and axions) and the cosmological

constant (Λ). Instead, it derives the observed gravitational anomalies in galaxies and cosmic expansion from a single geometric and logical foundation.

The Spacetime Drumbeat

CABM reinterprets our 4-dimensional universe as a boundary membrane (brane) embedded in a higher-dimensional space (bulk). Rather than being filled with invisible material particles, baryonic matter is modeled as standing-wave harmonics—essentially "cosmic sound"—generated by a fundamental bulk frequency (Ω_{bulk}) vibrating the spacetime membrane.

"We have spent decades and billions of dollars looking for dark matter particles in underground mines, finding absolutely nothing," says Chris

Numpsa. *"CABM shows that dark matter is not a substance. It is an ontological tension—a gravitational shadow cast by energy flowing from higher dimensions through micro-wormholes, combined with back-propagating echoes from the future state of the cosmos."*

A Causally Locked Universe

To handle the controversial concept of future states influencing the present (retrocausality), the framework applies a rigorous field-theoretic version of the *Novikov Self-Consistency Principle*. According to the authors, causal paradoxes are "topologically unrepresentable" in a continuous multidimensional field. The gravitational anchor holding galaxies together today is mathematically guaranteed to be the exact force required to shape the future structures emitting that very echo.

A Falsifiable Showdown with Current Cosmological Models

Unlike many hyper-dimensional theories that cannot be tested, CABM v0.2 delivers a definitive, brutal mathematical prediction for the upcoming International Pulsar Timing Array Data Release 3 (**IPTA DR3**).

While classical astrophysics attributes the cosmic gravitational wave background to Supermassive Black Hole Binaries (predicting a spectral strain tilt of $\beta = -2/3$), the CABM model proves that the quantum-topological flicker of the vibrating spacetime canvas will yield a tilt of exactly $\beta = -1/3$. Initial benchmarking against the landmark NANOGrav 15-year dataset shows that CABM's "Case C" prediction sits comfortably within 40–150% of observed limits,

outperforming traditional particle halos at high frequencies.

The manuscript has been released under an open-access **CC-BY-4.0** license and is available for public peer-review on Zenodo

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